PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA **Div: B** **Semester: III**	**Course Code: MCA01604** **Course Name: Data Science Laboratory**	**Batch: S2**
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CO No: CO605.1	**Assignment No: 01**	

Title: Use the Iris dataset and train a Logistic Regression model to classify whether a flower is of species Iris-setosa or not. Evaluate model accuracy, precision, and recall.

Code:

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, classification_report

# Load the Iris dataset
iris = load_iris()
X = iris.data
# Create binary target: 1 if setosa, 0 otherwise
Y = (iris.target == 0).astype(int)

# Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

# Train Logistic Regression model
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

# Predict on test set
y_pred = model.predict(X_test)

# Evaluate metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
report = classification_report(y_test, y_pred, target_names=['Not Setosa', 'Is Setosa'])

# Display the results
print("--- Model Evaluation for Iris-setosa Classification ---")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
```

```
print(f"Recall: {recall:.4f}")
print("\n--- Detailed Classification Report ---")
print(report)
```

Output:

```
--- Model Evaluation for Iris-setosa Classification ---
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000

--- Detailed Classification Report ---
              precision    recall  f1-score   support

  Not Setosa       1.00      1.00      1.00        20
   Is Setosa       1.00      1.00      1.00        10

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```